

# Bhanuja Karumuru

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## EDUCATION

**New York University**, MS in Computer Engineering New York, USA | Expected May 2027  
*Relevant Coursework:* High Performance Machine Learning, MLOps, Data Center & Cloud Computing, Internet Architecture

**NIT Sikkim**, B.Tech in Electronics and Communication Engineering Sikkim, India | Dec 2020 – May 2024  
*Relevant Coursework:* Data Structures & Algorithms, Machine Learning, Neural Networks/ Deep Learning, Computer Networks

## WORK EXPERIENCE

**Open-Source Contributor** *Github, Remote | Jul 2025 – Present*  
• Actively contributing and Merged upstream PRs in Conda improving CLI error handling and dependency resolution UX; changes adopted by core maintainers and shipped to production releases; continuing to contribute across multiple open-source repositories.

**Software Development Engineer, SalesUp - Yellowlake Technologies Services Private Limited** *Kolkata, India | July 2024 – July 2025*  
(Promoted from Software Engineering Intern)  
• Engineered Flask-based microservices for a high-traffic CRM platform; improved system throughput by 40% by refactoring synchronous REST endpoints into asynchronous task pipelines and optimizing database queries.  
• Architected an end-to-end generative AI feature for automated email campaigns using vector embeddings and LLMs - GPT APIs, increasing user engagement by 25%. Designed a RAG pipeline and implemented feedback loops to monitor hallucinations and improve audience segmentation.  
• Containerized using Docker and automated deployments with Jenkins, reducing release overhead by 30% and enabling zero-downtime rollouts.  
• Profiled AWS/Linux production environments optimized connection pooling and system-level configurations, reducing P99 API latency by 25%.

## RESEARCH EXPERIENCE

**AI/ML Infrastructure & Computer Vision Researcher, NYU RoboMaster (Team Ultraviolet)** *New York, USA | Jan 2026 – Present*  
• Developed and deployed real-time object detection, tracking, and environment understanding pipelines for autonomous robots within NYU's VIP RoboMaster program; built training and data processing pipelines for vision models and integrated perception outputs with controls and hardware teams for on-robot deployment.

**Graduate Research Contributor, New York University** *New York, USA | Sep 2025 – Present*  
• Researched virtualization-based OS security for ML workloads, implementing hypervisor-level privilege controls and kernel isolation to reduce cross-VM attack surfaces in multi-tenant training environments. Integrated namespace confinement and resource sandboxing primitives.

## PROJECTS

**FusedLinearAttention — Custom CUDA Kernel for Transformer Inference**  
• Designed a custom CUDA kernel fusing QKV projection and attention into a single pass, eliminating intermediate HBM roundtrips and reducing memory bandwidth consumption unavailable in FlashAttention-2 or NVIDIA Transformer Engine.  
• Validated numerical correctness against PyTorch reference across sequence lengths 64–1024 with tiled shared-memory management; profiled with NVIDIA NSight Systems to quantify HBM bandwidth savings, SM occupancy, and kernel launch overhead.  
• Benchmarked on a transformer forecasting model (ETTh1 dataset) on NYU Greene A100 nodes, demonstrating measurable HBM bandwidth and kernel overhead reduction over the unfused PyTorch baseline.

**Intelligent Deadline & Expiry Detection System — Production MLOps**  
• Built an end-to-end deadline extraction pipeline integrating a fine-tuned NER model (dslim/bert-base-NER) and a RoBERTa-based actionability classifier into Paperless-ngx, with a continuous retraining pipeline triggered by feedback accumulation or user correction rate.  
• Deployed on Chameleon with an asynchronous queue absorbing traffic bursts, auto-scaling to a second worker node on sustained queue depth, and production monitoring of inference latency, event creation volume, and correction rates.

**StyleSync — Large-Scale Recommendation System**  
• Built a two-tower deep retrieval recommender using TensorFlow Recommenders to learn user–item embeddings from large-scale interaction data, with FAISS-based ANN indexing pipelines achieving sub-100 ms end-to-end inference latency.  
• Productionized the system with containerized FastAPI services, scalable offline training pipelines, model versioning, and health checks, enabling reliable rollout of retrained models.

## TECHNICAL SKILLS

**Languages & Systems:** Python, C/C++, Java, Matlab, SQL, Bash, Linux, CUDA, Git

**ML & Data Science:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, FAISS, XGBoost, PEFT (LoRA/QLoRA), Pandas, NumPy, RAGAS

**Generative AI LLMs:** LangChain, OpenAI/GPT APIs, RAG Pipelines, Vector Databases, Prompt Engineering

**ML Systems & Infrastructure:** Docker, Kubernetes, MLflow, Jenkins, Model Serving, Monitoring, CI/CD

**Backend & Cloud:** FastAPI, Flask, REST APIs, AWS (EC2, S3, Lambda)

## PUBLICATIONS

**In-Domain Data Augmentation for Dysarthria Severity Classification, SPCOM 2024**  
Proposed in-domain augmentation techniques for dysarthric speech under low-resource conditions, improving severity classification accuracy.